Surface Phenomena & Catalysis
Week 3: Transport Limitations in Porous Catalysts

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2302337 — Semester 2/2025

Course Roadmap

Week	Chapter	Topic
1	Ch 1–2	Bridge Module + Computational Perspectives
2	Ch 3	Reactor Design & Catalyst Characterization
3	**Ch 4**	**Transport Limitations in Porous Catalysts**
4	Ch 5	Selectivity Engineering
5	Ch 6	Zeolites & Carbon Nanotubes

If the pellet is too large, how much activity do we lose to diffusion?

Today's Agenda

Part I: Fundamentals *~30 min*

Effective diffusivity, shell balance, Thiele modulus, effectiveness factor, concentration profiles

Part II: Diagnostics & Applications *~35 min*

Weisz–Prater criterion, diffusion regimes, external mass transfer, reactor design with η

Part III: Summary & Self-Study *~10 min*

Key equations + NB4 self-study guide + looking ahead to Week 4

Lecture notes: Chapter 4 — Notebook: NB4

Outline

1 Part I: Fundamentals

2 Part II: Diagnostics & Applications

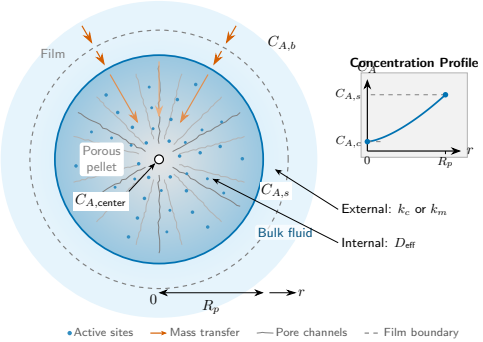
3 Summary & Self-Study

Why Transport Matters

In Week 2, we assumed the intrinsic rate law governs reactor performance. But what if reactant cannot reach the active sites fast enough?

The problem:

- Porous catalysts have **internal** active sites
- Reactant must diffuse **through pores** to reach them
- If reaction is fast relative to diffusion, the pellet interior is "starved"



Consequence

Observed rate $<$ intrinsic rate.
The catalyst is underutilized.

External vs Internal Mass Transfer

External (film) resistance

- Stagnant boundary layer around pellet
- Concentration drop: $C_{A,b} \to C_{A,s}$
- Governed by flow conditions
- Diagnostic: vary flow rate at constant τ

Internal (pore) resistance

- Diffusion through tortuous pore network
- Concentration drop: $C_{A,s} \to C_A(r)$
- Governed by pellet geometry & porosity
- Diagnostic: vary pellet size (crushing test)

Today's focus: internal resistance first (dominant in most gas-phase systems), then external, then both.

Effective Diffusivity

Diffusion inside a porous pellet is slower than in free space:

- Only a fraction is pore space (porosity ε)
- Pores are not straight (tortuosity τ_p)

$$D_{\text{eff}} = \frac{\varepsilon}{\tau_p} D_{\text{pore}}$$

Typical values for gases in mesoporous catalysts: $D_{\text{eff}} \approx 10^{-7}$–$10^{-6}$ m^2/s

Typical parameters

Property	Value
Porosity ε	0.3–0.6
Tortuosity τ_p	3–5
D_{pore} (gas)	10^{-6}–10^{-5} m^2/s
D_{pore} (liquid)	10^{-10}–10^{-9} m^2/s

Rule of thumb

$D_{\text{eff}} \approx D_{\text{pore}}/10$

(assumes $\varepsilon = 0.5$, $\tau_p = 4$)

Shell Balance: Reaction–Diffusion Equation

Steady-state mass balance on a thin spherical shell:

Step 1: Fick's first law: $J_A = -D_{\text{eff}}\,\mathrm{d}C_A/\mathrm{d}r$

Step 2: Balance diffusion in = reaction consumed

Step 3: Take $\mathrm{d}r \to 0$:

$$D_{\text{eff}}\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}C_A}{\mathrm{d}r}\right) = r(C_A)$$

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Boundary conditions:

1. Symmetry: $\mathrm{d}C_A/\mathrm{d}r\big|_{r=0} = 0$
2. Surface: $C_A(R_p) = C_{A,s}$

For 1st-order kinetics

Substitute $r = k\,C_A$:

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}C_A}{\mathrm{d}r}\right) = \frac{k}{D_{\text{eff}}}C_A$$

Substitution $u = r\,C_A$ transforms this into a simple 2nd-order ODE with $\sinh/\cosh$ solutions.

The Thiele Modulus

$$\phi = R_p \sqrt{\frac{k}{D_{\text{eff}}}}$$

(sphere, 1st order)

Physical meaning:
ϕ^2 = reaction rate/diffusion rate

- $\phi \ll 1$: Diffusion is fast. Uniform C_A throughout pellet.
- $\phi \gg 1$: Reaction is fast. Pellet center is "starved."

Three regimes

ϕ	**Regime**
< 0.3	Kinetic control
0.3–3	Transition
> 3	Diffusion limited

Slab geometry

$\phi_{\text{slab}} = L\sqrt{k/D_{\text{eff}}}$
(L = half-thickness)

Concept Check — Thiele Modulus

Concept Check

1. A reaction has $k = 10$ s^{-1}, $D_{\text{eff}} = 2 \times 10^{-6}$ m^2/s, and $R_p = 2$ mm. Calculate ϕ and predict the regime.

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1. A reaction has $k = 10$ s^{-1}, $D_{\text{eff}} = 2 \times 10^{-6}$ m^2/s, and $R_p = 2$ mm. Calculate ϕ and predict the regime. $\phi = 0.002 \times \sqrt{10/(2 \times 10^{-6})} = 0.002 \times 2236 \approx \mathbf{4.5}$ (diffusion limited)

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3. A liquid-phase and gas-phase reaction have the same k and R_p. Which is more likely to be diffusion-limited? Why?

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3. A liquid-phase and gas-phase reaction have the same k and R_p. Which is more likely to be diffusion-limited? Why? **Liquid-phase**: D_{eff} is $\sim 10^4\times$ smaller $\Rightarrow$ much larger ϕ

The Effectiveness Factor

$$\eta \equiv \frac{\text{actual rate in porous pellet}}{\text{rate if entire pellet at } C_{A,s}}$$

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Practical payoff:

$$r_{\text{obs}} = \eta \cdot r_{\text{intrinsic}}(C_{A,s})$$

- $\eta = 1$: No diffusion limitation. Full pellet utilization.
- $\eta < 1$: Part of the pellet is "wasted" due to concentration depletion.
- $\eta > 1$: Possible for exothermic reactions (non-isothermal).

The key idea

Once we know η, we use the **intrinsic** rate law with a simple correction factor. All reactor design equations from Week 2 still apply — just replace k with $\eta \cdot k$.

Effectiveness Factor: Slab & Sphere

Flat slab (half-thickness L):

$$\eta_{\text{slab}} = \frac{\tanh \phi}{\phi}$$

Sphere (radius R_p):

$$\eta_{\text{sphere}} = \frac{3}{\phi^2}\left(\phi \coth \phi - 1\right)$$

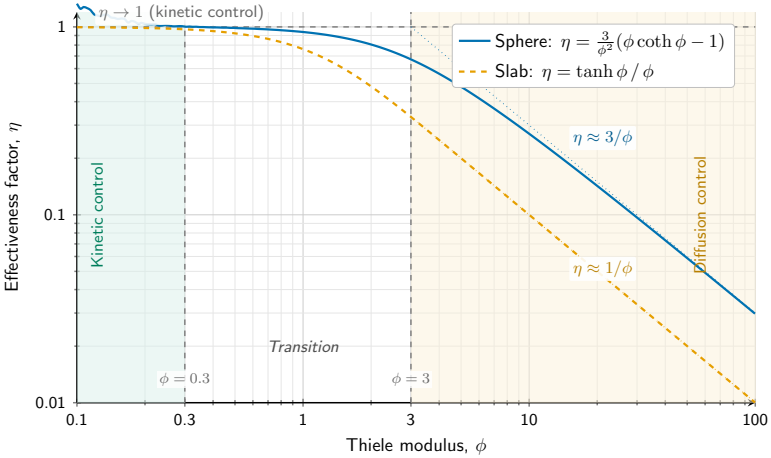
Asymptotic limits

	$\phi \ll 1$	$\phi \gg 1$
Slab	$\eta \to 1$	$\eta \to 1/\phi$
Sphere	$\eta \to 1$	$\eta \to 3/\phi$

Aris characteristic length

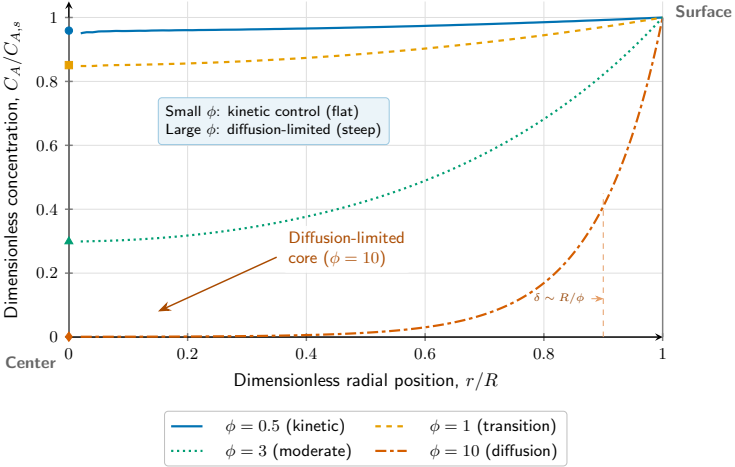
Using $L_c = V_p/S_{\text{ext}}$ ($R_p/3$ for sphere), the curves collapse to within ~10%.

Effectiveness Factor vs Thiele Modulus



Three regimes: kinetic control ($\phi < 0.3$, $\eta \approx 1$) — transition ($0.3 < \phi < 3$) — diffusion limited ($\phi > 3$, $\eta \propto 1/\phi$)

Concentration Profiles Inside a Spherical Pellet



$\phi = 0.5$: Nearly uniform — entire pellet active. $\phi = 10$: Depleted at $r/R_p < 0.8$; penetration depth $\delta \sim R_p/\phi$.

Apparent Activation Energy Halving

If $r_{obs} \propto \sqrt{k}$ instead of k, what happens to the Arrhenius slope?

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When $\phi \gg 1$ (diffusion-limited regime):

$$r_{\text{obs}} \approx \frac{3}{R_p} \sqrt{D_{\text{eff}} \cdot k}\, C_{A,s}$$

Since $k = A\, e^{-E_{\text{a}}/RT}$ and D_{eff} has weak T-dependence:

$$r_{\text{obs}} \propto \sqrt{e^{-E_{\text{a}}/RT}} = e^{-E_{\text{a}}/(2RT)}$$

$$E_{\text{app}} \approx \frac{E_{\text{a}}}{2}$$

The observed Arrhenius slope is halved because r_{obs} depends on $\sqrt{k}$, not k.

$E_{\text{app}} \approx E_{\text{a}}/2$: Diagnostic & Caveat

Diagnostic significance

- If Arrhenius plot gives $E_{\text{app}} \approx E_{\text{a}}/2 \Rightarrow$ diffusion is limiting
- Crush the pellet: if full E_{a} is restored, diffusion was the culprit
- This is one of the most powerful **experimental diagnostics** for internal transport limitations

Caveat: micropores

For activated diffusion (zeolites):

$$E_{\text{app}} \approx \frac{E_{\text{a}} + E_D}{2}$$

When E_D is large, the halving is **masked** — E_{app} may be close to E_{a}!

Use the Weisz–Prater criterion instead.

Diffusion type	$D_{\text{eff}}(T)$	E_{app}
Non-activated (mesopores)	$\propto T^{0.5\text{–}1.75}$	$\approx E_{\text{a}}/2$
Activated (micropores)	$\propto e^{-E_D/RT}$	$\approx (E_{\text{a}} + E_D)/2$

Worked Example: Pellet Size Effect on η

Problem

A first-order reaction: $k = 1.0$ s^{-1}, $D_{\text{eff}} = 1.0 \times 10^{-6}$ m^2/s. Calculate η for spheres with $R_p = 0.1$, 1.0, 5.0 mm. *Predict: which will have $\eta \approx 1$?*

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First: $\sqrt{k/D_{\text{eff}}} = \sqrt{10^6} = 10^3$ m^{-1}

R_p (mm)	R_p (m)	$\phi = R_p \times 10^3$	η (exact)	$3/\phi$	Regime
0.1	10^{-4}	0.1	**0.999**	—	Kinetic control
1.0	10^{-3}	1.0	**0.94**	3.0	Transition
5.0	5×10^{-3}	5.0	**0.48**	0.60	Diffusion limited

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Key insight

A 50-fold increase in R_p (0.1 $\rightarrow$ 5 mm) reduces utilization from ~100% to 48%. This is why industrial catalysts use small pellets or thin washcoats on monolith supports.

Concept Check — Effectiveness Factor

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2. You reduce R_p by a factor of 4. What is the new ϕ? Is it still diffusion-limited?

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$\phi_{\text{new}} = 20/4 = \mathbf{5}$; still diffusion-limited ($\phi > 3$)

From Theory to Practice

What we have so far

- The Thiele modulus ϕ captures the competition between reaction and diffusion
- The effectiveness factor $\eta(\phi)$ quantifies how much rate is lost
- Pellet size, intrinsic k, and D_{eff} all enter through ϕ

The practical problem

We measure r_{obs} but do **not** know intrinsic k or η separately. How do we diagnose whether diffusion is limiting using **observable** quantities?

The Weisz–Prater criterion provides the answer.

Next: diagnostics, diffusion regimes, and reactor design with η.

Outline

1 Part I: Fundamentals

2 Part II: Diagnostics & Applications

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The Weisz–Prater Criterion

Key idea: a diagnostic based entirely on **observable** quantities.

$$N_{WP} = \frac{r_{\text{obs}} \cdot R_p^2}{C_{A,s} \cdot D_{\text{eff}}}$$

Connection to ϕ and η:

$$N_{WP} = \eta \cdot \phi^2$$

- All quantities on the RHS are measurable (r_{obs}, R_p, $C_{A,s}$, D_{eff})
- No need to know intrinsic k

Interpretation

- $N_{WP} \ll 1$: Internal diffusion **negligible**
- $N_{WP} \sim 1$: **Moderate** limitation
- $N_{WP} \gg 1$: **Severe** diffusion limitation

Weisz–Prater Thresholds

For $\eta > 0.95$ (less than 5% rate loss to diffusion):

Order n	**Criterion**	**Physical reason**
0 (zero)	$N_{WP} < 6$	Rate independent of C_A; diffusion matters only when $C_A \to 0$ (dead core)
1 (first)	$N_{WP} < 0.3$	Benchmark case; most widely used
2 (second)	$N_{WP} < 0.15$	Rate drops as C_A^2; more sensitive to internal gradients

Why do thresholds differ?

Higher-order reactions are more sensitive to concentration drops. Zero-order reactions are insensitive until complete depletion — hence the very generous threshold.

The Crushing Test: Direct Experimental Diagnostic

The most direct test for internal diffusion — **no knowledge of D_{eff} required.**

Procedure:

1. Measure r_{obs} with original pellets
2. Crush to a smaller particle size
3. Measure r_{obs} again

Interpretation

- r_{obs} **unchanged**:
  Kinetic control ($\phi < 0.3$)
- $r_{\text{obs}} \propto 1/R_p$:
  Diffusion limited ($\phi > 3$)
- Rate increases but $\neq 1/R_p$:
  Transition regime

Why $r_{\text{obs}} \propto 1/R_p$?

In the diffusion-limited regime:

$$\eta \approx \frac{3}{\phi} \propto \frac{1}{R_p}$$

$$r_{\text{obs}} = \eta \cdot k\, C_{A,s} \propto \frac{1}{R_p}$$

Complementary tools

WP criterion: *could* diffusion matter?
Crushing test: *does* diffusion matter?

Worked Example: Weisz–Prater Diagnostic

Problem

Observed rate: $r_{\text{obs}} = 0.05$ mol/(L·s), $R_p = 2$ mm, $C_{A,s} = 0.5$ mol/L, $D_{\text{eff}} = 5 \times 10^{-7}$ m^2/s.
Is internal diffusion significant? *Predict: N_{WP} above or below 0.3?*

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Convert units: $r_{\text{obs}} = 50$ mol/(m^3·s), $R_p = 2 \times 10^{-3}$ m, $C_{A,s} = 500$ mol/m^3

$$N_{WP} = \frac{50 \times (2 \times 10^{-3})^2}{500 \times 5 \times 10^{-7}} = \frac{50 \times 4 \times 10^{-6}}{2.5 \times 10^{-4}} = \frac{2 \times 10^{-4}}{2.5 \times 10^{-4}} = \mathbf{0.8}$$

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Diagnosis

$N_{WP} = 0.8 > 0.3$
$\Rightarrow$ Internal diffusion is **moderately significant**.

Fix: reduce pellet size

At $R_p = 0.5$ mm:
$N_{WP} = 0.8 \times (0.5/2)^2 = \mathbf{0.05}$
($N_{WP} \propto R_p^2$: 4× smaller $R_p \rightarrow$ 16× smaller N_{WP})

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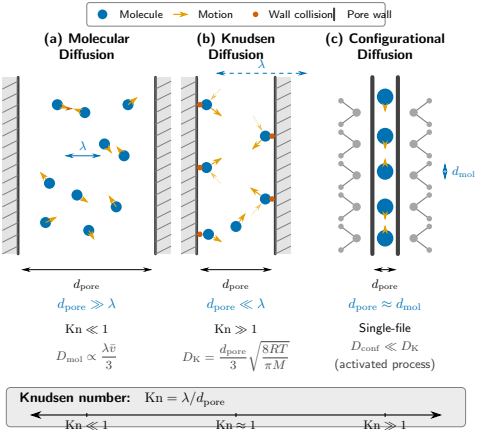
2. A reaction has $N_{WP} = 0.8$ (transition regime). A colleague suggests switching from 2 mm to 0.5 mm pellets. Predict the new N_{WP}. $N_{WP} \propto R_p^2$: $0.8 \times (0.5/2)^2 = \mathbf{0.05} \ll 0.3$ (kinetic control)

Three Diffusion Regimes in Porous Catalysts



Molecular ($d_{\text{pore}} \gg \lambda$): $D \sim 10^{-5}$–10^{-4} m^2/s — **Knudsen** ($d_{\text{pore}} \ll \lambda$): $D_K \sim 10^{-7}$–10^{-5} m^2/s — **Configurational** ($d_{\text{pore}} \approx d_{\text{mol}}$): $D \sim 10^{-14}$–10^{-8} m^2/s

Knudsen Diffusion

When $d_{\text{pore}} \ll \lambda$ (mean free path), molecules collide with **walls** more than with each other.

$$D_K = \frac{2}{3} r_p \sqrt{\frac{8RT}{\pi M}} \propto r_p \sqrt{\frac{T}{M}}$$

Key dependences:

- $D_K \propto r_p$: wider pores $\rightarrow$ faster
- $D_K \propto \sqrt{T}$: mild T dependence
- $D_K \propto 1/\sqrt{M}$: light molecules diffuse faster
- **Independent of pressure** (unlike molecular diffusion)

Quick calculation: CO_2 at 500 K

$r_p = 5$ nm, $M = 44$ g/mol:

$$D_K \approx 1.6 \times 10^{-6} \text{ m}^2/\text{s}$$

About 6× smaller than free-space D_{AB}.
Nanoscale pores significantly reduce transport.

Microporous Transport: Zeolites vs Nanotubes

Zeolites: configurational diffusion

- Pore apertures: 0.3–1.2 nm
- $D \sim 10^{-14}$–10^{-8} m^2/s
- **Activated** process: $D = D_0\, e^{-E_D/RT}$
- E_D depends on molecule/pore fit
- Large ϕ even for small crystals

Three "transport multipliers": small pores, strong adsorption, shape selectivity.

Carbon nanotubes: enhanced Knudsen

- Inner diameter: 1–10 nm
- Smooth, graphitic walls
- $D_{\text{CNT}} = F \cdot D_K$
- Enhancement: $F = 10$–$10{,}000$
- Specular reflection $\neq$ diffuse scattering

CNTs can **overcome** the Knudsen penalty that limits zeolites. See Chapter 6.

Same pore size, dramatically different transport — wall chemistry matters!

External Mass Transfer

Biot number for mass transfer (k_c = external mass transfer coeff., m/s):

$$Bi_m = \frac{k_c R_p}{D_{\text{eff}}}$$

- $Bi_m \gg 1$: External transfer fast; **internal** diffusion controls
- $Bi_m \ll 1$: External transfer slow; **film** resistance controls

Sherwood correlation (packed bed):

$$Sh = \frac{k_c d_p}{D_{AB}} = 2 + 1.1\, Re^{0.6}\, Sc^{1/3}$$

Gas vs liquid

	Gas	**Liquid**
D_{AB} (m^2/s)	10^{-5}	10^{-9}
Sc	$\sim$1	$\sim$1000
k_c	Large	Small
External limitation?	Rarely limits	Often limits

Diagnostic

Increase flow rate at constant τ.
If rate changes $\Rightarrow$ external limitation.

Overall Effectiveness Factor

Combining internal and external resistances (sphere):

$$\eta_{\text{overall}} = \frac{\eta}{1 + \eta\,\phi^2/(3\,Bi_m)} \quad \text{(sphere)}$$

Three limiting cases:

1. $3\,Bi_m \gg \eta\phi^2$:
   $\eta_{\text{ov}} \to \eta$ (internal control)
2. $3\,Bi_m \ll \eta\phi^2$:
   $\eta_{\text{ov}} \to 3\,Bi_m/\phi^2$ (external control)
3. $3\,Bi_m \sim \eta\phi^2$:
   Both resistances matter

Factor of 3

The factor of 3 comes from the sphere volume-to-surface ratio: $V_p/S_{\text{ext}} = R_p/3$.

For a slab: no factor of 3; replace $\phi^2/(3\,Bi_m)$ with $\phi^2_{\text{slab}}/Bi_{m,\text{slab}}$.

Reactor Design with Effectiveness Factor

Replace intrinsic k with effective rate constant $k_{\text{eff}} = \eta \cdot k$:

PFR (first-order):

$$X = 1 - \exp(-\eta\, k\, \tau)$$

CSTR (first-order):

$$X = \frac{\eta\, k\, \tau}{1 + \eta\, k\, \tau}$$

Practical impact

- Diffusion reduces the **effective** rate constant
- Reactor must be **larger** to achieve the same conversion
- Cost scales with $\tau \propto 1/\eta$

Temperature complication

η varies with T through $\phi(T)$:
$k(T)$ rises fast, $D_{\text{eff}}(T)$ rises slowly
$\Rightarrow \phi$ increases with T
$\Rightarrow \eta$ decreases with T

Worked Example: PFR Conversion with Diffusion Limitation

Problem

PFR, first-order kinetics: $k = 0.5$ s^{-1}, $\tau = 4$ s. Compare conversion with $\eta = 1$ vs $\eta = 0.4$.

Predict: will the conversion drop by more or less than half?

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No limitation $(\eta = 1)$:

$$k_{\text{eff}} = 0.5 \text{ s}^{-1}$$
$$X = 1 - e^{-0.5\times 4}$$
$$= 1 - e^{-2} = \mathbf{86.5}\%$$

With limitation $(\eta = 0.4)$:

$$k_{\text{eff}} = 0.4 \times 0.5 = 0.2 \text{ s}^{-1}$$
$$X = 1 - e^{-0.2\times 4}$$
$$= 1 - e^{-0.8} = \mathbf{55.1}\%$$

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With limitation ($\eta = 0.4$):

$$k_{\text{eff}} = 0.4 \times 0.5 = 0.2 \text{ s}^{-1}$$
$$X = 1 - e^{-0.2\times 4}$$
$$= 1 - e^{-0.8} = \mathbf{55.1}\%$$

Key result

Conversion drops by **31 percentage points**. To recover 86.5% conversion, increase τ from 4 s to 10 s — a **2.5× larger reactor**.

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3. Two catalysts have the same k and crystal radius: a zeolite ($D \sim 10^{-12}$ m^2/s) and a CNT support ($D \sim 10^{-6}$ m^2/s). Which has a higher ϕ? **Zeolite**: D is $10^6\times$ smaller $\Rightarrow$ ϕ is $10^3\times$ larger

Pellet Size Optimization

Small pellets:

+ Low ϕ, high η
+ Better utilization of catalyst
− High pressure drop ($\Delta P \propto 1/d_p^2$)
− Difficult to separate from product

Large pellets:

+ Low pressure drop
+ Easy handling
− High ϕ, low η
− Wasted catalyst interior

Engineering solutions

- **Eggshell catalysts:** Active phase deposited only near surface
- **Monoliths:** Thin washcoat ($\sim\mu$m) on structured support
- **Shaped pellets:** Trilobes, rings — increase S_{ext}/V_p

Design rule

Choose R_p such that $\phi \lesssim 1$ (or use structured geometries to decouple ΔP from η).

Industrial Case: Monolith Catalysts Revisited

Callback to Week 2: automotive three-way catalyst

- Washcoat thickness: $\sim$20–50 μm
- At $L = 25$ μm and typical D_{eff}, k: $\phi \ll 1 \Rightarrow \eta \approx 1$
- The monolith design **eliminates** the diffusion penalty while maintaining low ΔP

Why it works

Thin washcoat = small L_c.
Open channels = low pressure drop.
Best of both worlds.

Counterexample: Fischer–Tropsch

- $CO + H_2 \rightarrow$ hydrocarbons (Co or Fe catalyst)
- Large pellets ($\sim$mm) used in packed beds
- $\phi \sim 3$–10 (intentionally!)
- In liquid-filled pores, diffusion depletes H_2 relative to CO $\rightarrow$ favours chain growth $\rightarrow$ heavier products (C_{5+})
- $\eta < 1$ is a **feature**, not a bug

Transport Limitations: A Complete Framework

Question	Tool	Criterion
Is internal diffusion significant?	Weisz–Prater	$N_{WP} < 0.3$ (1st order)
Is external transfer significant?	Vary flow rate	Rate changes?
What diffusion regime?	Knudsen number	$Kn = \lambda / d_{\text{pore}}$
What is the rate penalty?	$\eta(\phi)$	$r_{\text{obs}} = \eta \cdot r_{\text{intr}}$
How to fix it?	Reduce R_p, use structured catalyst	$\phi \propto R_p$

The transport hierarchy

$$C_{A,b} \xrightarrow{\text{external film}} C_{A,s} \xrightarrow{\text{pore diffusion}} C_A(r) \xrightarrow{\text{reaction}} \text{products}$$

Each step can be rate-limiting. Identify the bottleneck before designing solutions.

Outline

1 Part I: Fundamentals

2 Part II: Diagnostics & Applications

3 Summary & Self-Study

Key Equations Card

Transport Fundamentals:

1. D_{eff}**:** $D_{\text{eff}} = (\varepsilon/\tau_p)\, D_{\text{pore}}$
2. **Shell balance:** $D_{\text{eff}} \frac{1}{r^2} \frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 \frac{\mathrm{d}C_A}{\mathrm{d}r}\right) = r(C_A)$
3. **Thiele:** $\phi = R_p \sqrt{k/D_{\text{eff}}}$
4. η **(slab):** $\eta = \tanh\phi/\phi$
5. η **(sphere):** $\eta = \frac{3}{\phi^2}(\phi \coth\phi - 1)$

Diagnostics & Design:

6. **Weisz–Prater:** $N_{WP} = r_{\text{obs}}\, R_p^2/(C_{A,s}\, D_{\text{eff}})$
7. **Knudsen:** $D_K = \frac{2}{3}\, r_p \sqrt{8RT/(\pi M)}$
8. **Overall** η**:** $\eta_{\text{ov}} = \eta/(1 + \eta\,\phi^2/(3\,Bi_m))$
9. **PFR with** η**:** $X = 1 - \exp(-\eta\, k\, \tau)$
10. E_{app} **halving:** $E_{\text{app}} \approx E_{\text{a}}/2$ when $\phi \gg 1$

Ten equations — from pellet physics to reactor sizing with diffusion.

NB4: Self-Study Exercises

Exercises (complete before Week 4)

Ex	Topic	Priority
1	η vs ϕ plotting	Essential
2	Pellet size sensitivity	Essential
3	Weisz–Prater diagnosis	Essential
4	Knudsen vs molecular D	Essential
5	PFR conversion with η	Bonus

35 cells | ∼2–3 h |
`https://github.com/ExaPsi/2302337`

Self-study from Chapter 4

Topics not covered in today's lecture:

- Generalized ϕ for nth-order (§4.4)
- Aris characteristic length (§4.4)
- Non-isothermal pellets (§4.9)
- Activated diffusion in zeolites (§4.7)
- Mears criterion for external transfer (§4.8)

Notebook roadmap

Week 1–2	NB1, NB2, NB3
Week 3	**NB4**
Week 4–5	NB5, NB6

Looking Ahead — Week 4

Next: Selectivity Engineering (Chapter 5)

- Parallel vs consecutive reaction networks
- Optimal space time τ^* for maximum yield of intermediate
- CSTR vs PFR selectivity differences

Preparation

- Read Chapter 5, Sections 5.1–5.3
- Complete NB4 essential exercises
- Review PFR/CSTR design equations (Week 2)

Key question for Week 4

When multiple products can form, how do we choose reactor conditions to maximize the **desired product**?

See you next week!